

Title

Dose-Response and Bioenhancer Analysis in Turmeric-Based Nutritional Pain Management

Abstract

Secondary analysis examined pain response patterns across different formulations within the crossover trial. Participants exposed to varying concentrations of turmeric showed similar reductions in reported pain scores regardless of dose level. When black pepper was added as a proposed bioenhancer, participants did not demonstrate substantially greater pain reduction compared to turmeric-only periods. Subgroup analysis considering baseline pain severity, age stratification, and prior medication use revealed no significant interactions with treatment response. Adherence to the daily reporting protocol remained high across all study arms. Long-term effects beyond the 21-day window could not be assessed in this preliminary design. These findings suggest that the analgesic signal observed with turmeric does not exhibit clear dose-response relationships within the culinary range tested, and the theoretical bioenhancing effect of pepper adjunction did not translate to measurable clinical advantage in this cohort. Recommendations for next-stage investigation include extended follow-up periods and evaluation in distinct chronic pain populations.